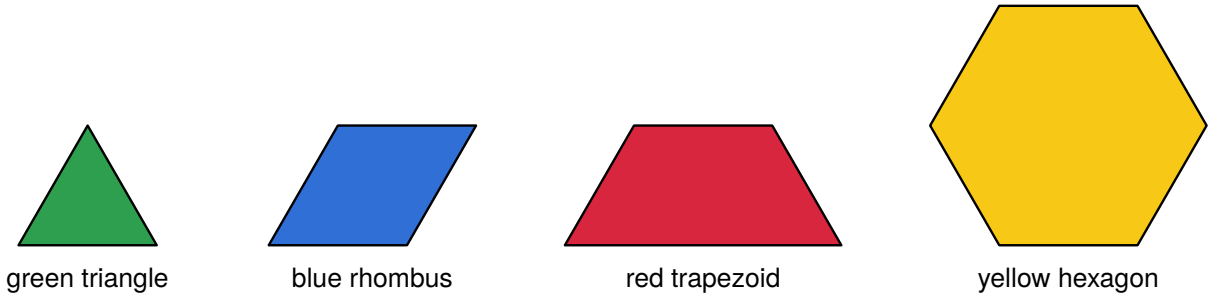
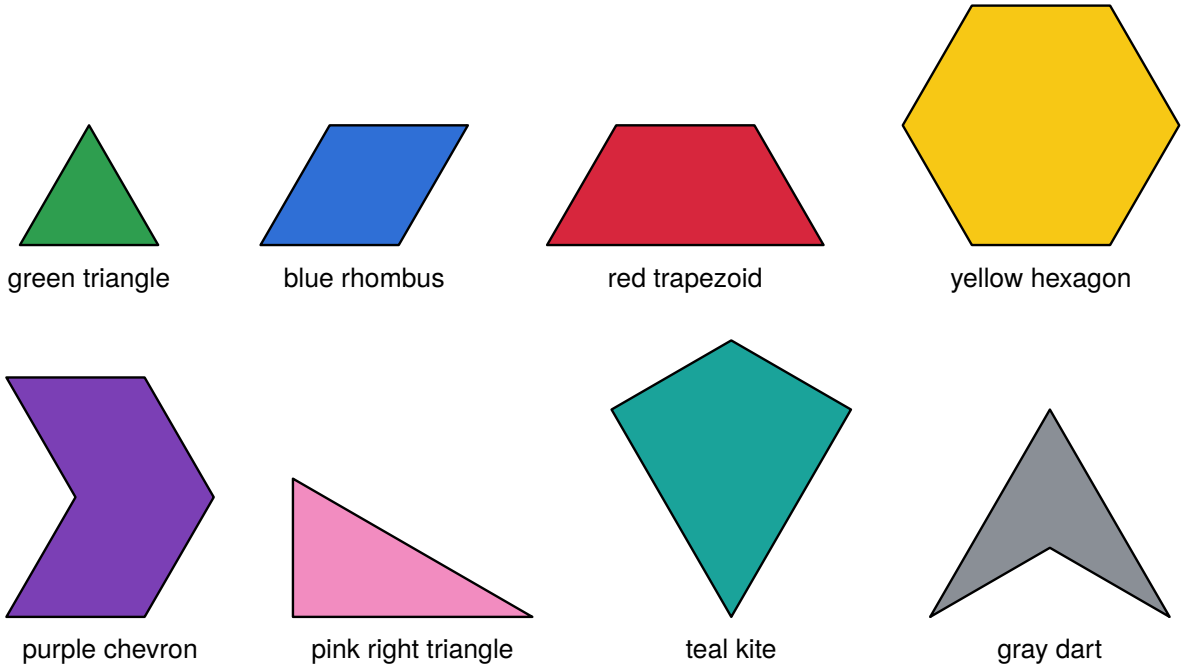
Problem 1: Your partner picks one of these four blocks in secret. Can you always find out which one with two yes-or-no questions?



green triangle | blue rhombus | red trapezoid | yellow hexagon

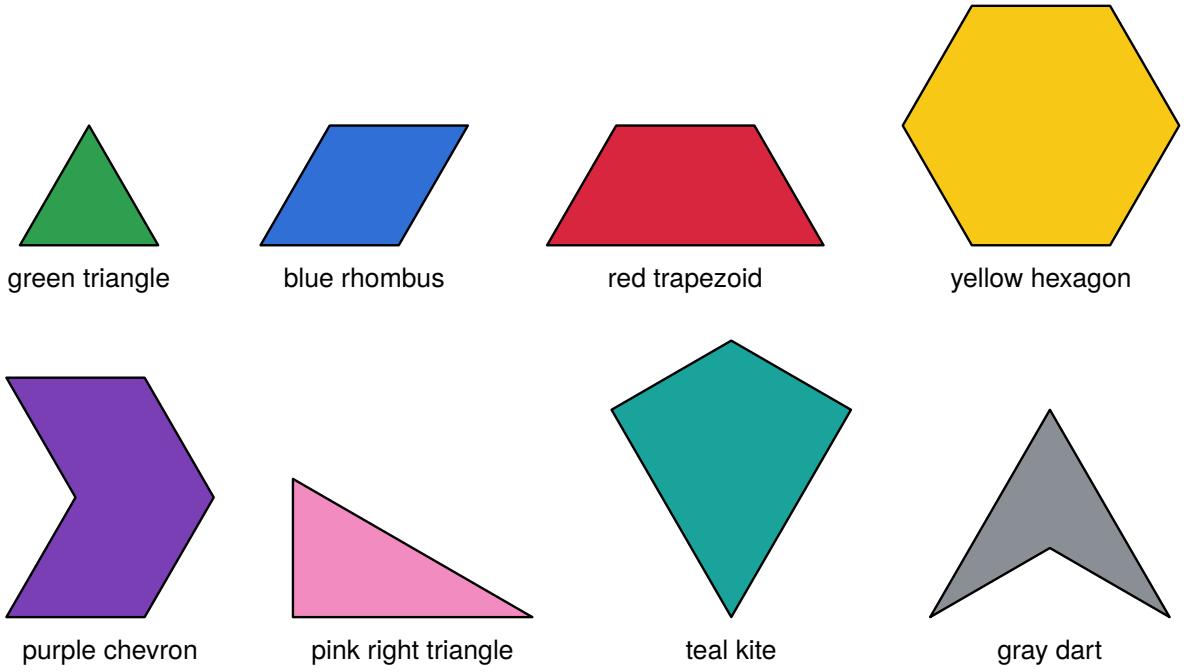
Problem 2: Your partner picks one of these eight blocks in secret. You can only point and ask “Is it this one?” What is the most questions you could need before you are sure?



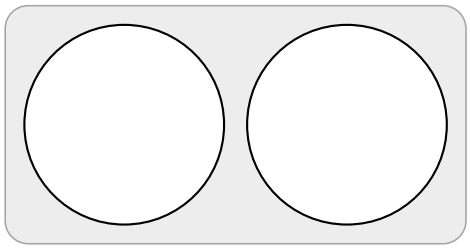
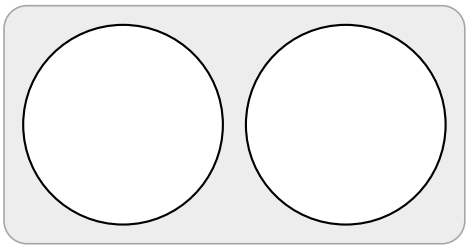
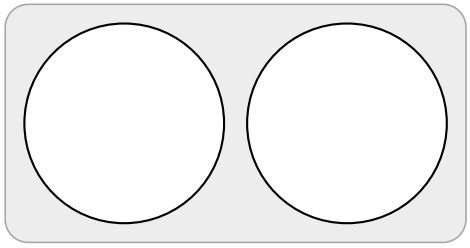
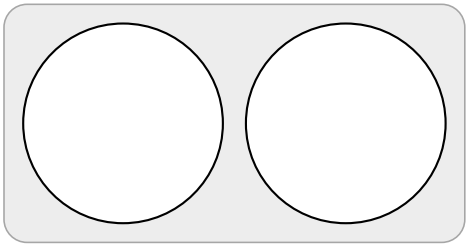
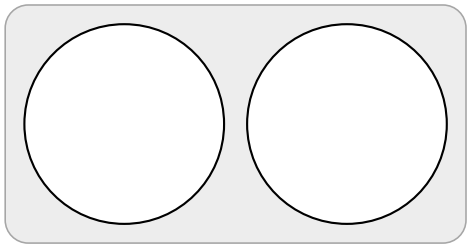
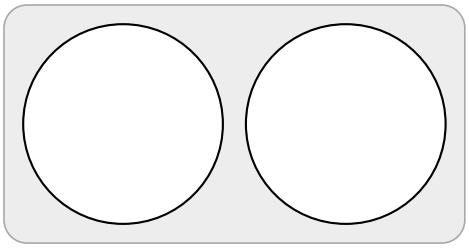
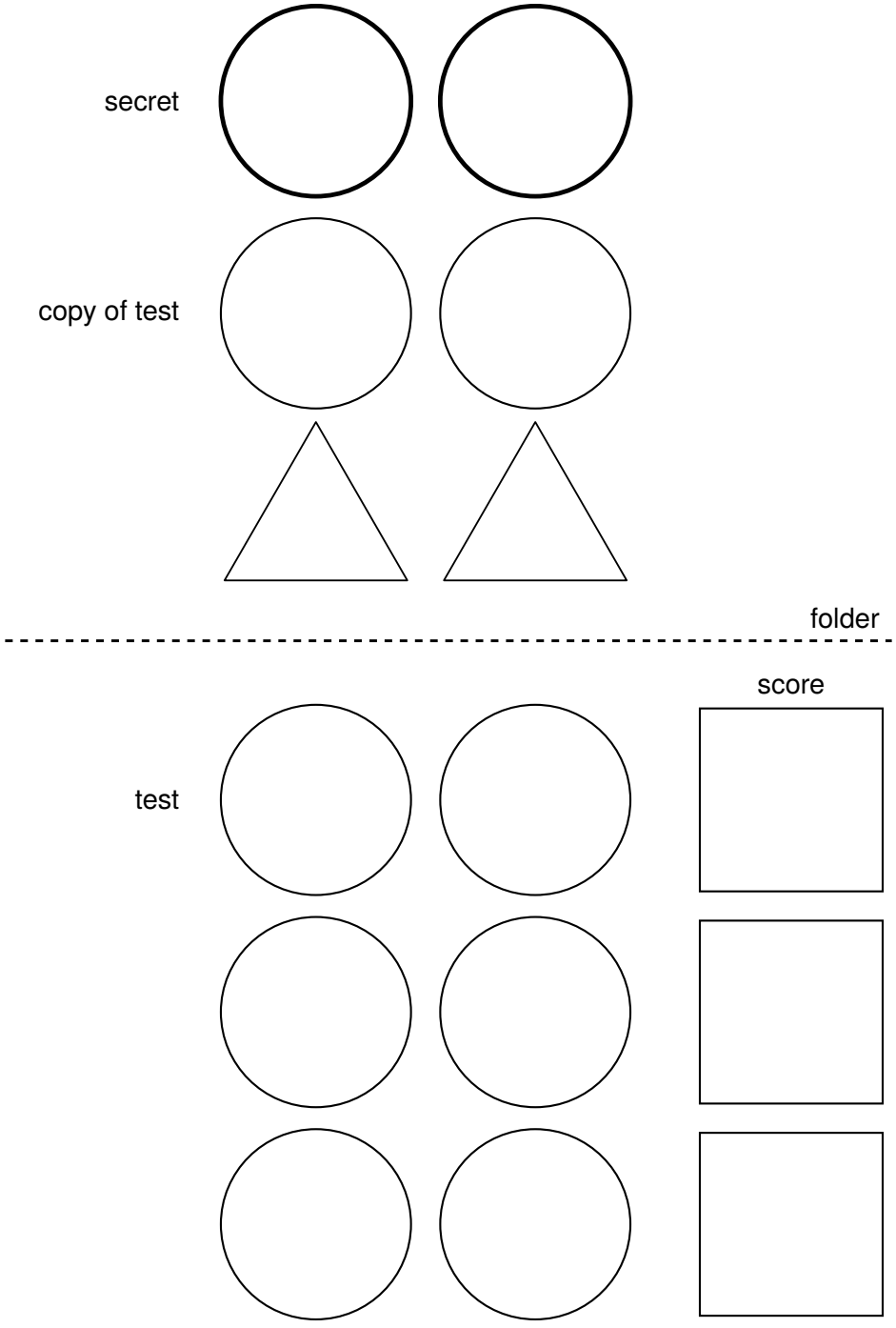
green triangle | blue rhombus | red trapezoid | yellow hexagon

purple chevron | pink right triangle | teal kite | gray dart

Problem 3: Your partner picks one of these eight blocks in secret. Can you always find out which one with three yes-or-no questions?



Problem 5: A secret is two counters in a row, each one red or yellow. How many different secrets can you make?








Problem 6: The grown-up hides two counters behind the folder. Make a test with two counters, and the grown-up says how many match. Can you always be sure of the secret after one test? After two tests?